INCH-POUND
MIL-PRF-32002B
18 September 1998
SUPERSEDING
MIL-PRF-32002A
5 September 1997

PERFORMANCE SPECIFICATION

COVER MATERIAL, VEHICLE, COATED TACTICAL AND COMBAT, HEAVY-DUTY, WATERPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for heavy-duty, waterproof, coated tactical and combat vehicle cover material, known hereinafter as "cover material" (see 6.1).

1.2 Classification. The heavy-duty, waterproof, cover material is of the classes listed below:

Class A	- Dual sided infrared protection/low gloss
Class B	- Single sided infrared protection/low gloss.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 25GP

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods (See 4.4.1).
FED-STD-595	- Colors Used in Government Procurement.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

49 CFR, Section 571.302	- Code of Federal Regulations (CFR).

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR, Part 798	- Code of Federal Regulations (CFR).

(Copies of the Code of Federal Regulations are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1 - Gray Scale for Color Change.

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, 1 Davis Drive, PO Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D471	- Standard Test Method for Rubber Property - Effect of Liquids (DoD Adopted).
ASTM D523	- Standard Test Method for Specular Gloss (DoD Adopted).
ASTM D751	- Standard Test Methods for Coated Fabrics (DoD Adopted).
ASTM D2136	- Standard Test Method for Coated Fabrics - Low-Temperature Bend Test (DoD Adopted).
ASTM D3389	- Standard Test Method for Coated Fabrics Abrasion Resistance (Rotary Platform, Double-Head Abrader) (DoD Adopted).
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test) (DoD Adopted).
ASTM D5035	- Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method) (DoD Adopted).
ASTM G21	- Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi (DoD Adopted).
ASTM G26	- Standard Practice for Operating Light-Exposure Apparatus (Xenon-Arc Type) with and without Water for Exposure of Nonmetallic Materials (DoD Adopted).

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(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J883	- Test Method for Determining Dimensional Stability of Automotive Textile Materials.
SAE J1351	- Hot Odor Test for Insulation Materials.
SAE J1966	- Lubricating Oil, Aircraft Piston Engine (Nondispersant Mineral Oil) (DoD Adopted).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

TAPPI

TAPPI T451 CM	- Flexural Properties of Paper (Clark Stiffness).

(Application for copies should be addressed to TAPPI, 15 Technology Parkway S, Norcross, GA 30092.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. The base cloth of the cover material shall be plain woven with high-tenacity yarn and a minimum of 25 yarns per inch in both the warp and fill direction. Natural fibers shall not be used (see 6.5). The base cloth shall be coated with poly (vinyl chloride) (PVC), polyurethane or their equivalent (see 6.6).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.3 Operating requirements.

3.3.1 Infrared reflectance. The cover material shall meet the required infrared reflectance properties as specified in table I for both sides (Class A) of the cover material or one side only (Class B) (see 6.2). The cover material shall not be painted with chemical agent resistant coating (CARC) paint (see 4.4.3.1).

TABLE I. Infrared reflectance requirements.

Color 1/	Infrared 2/		Allowable 3/
	Min.	Max.	Ratio min.
Dark green 34082	-	60.0	5.2
Green 383, 34094	-	60.0	5.2
Field drab 33105	25.0	35.0	-
Earth yellow 33245	30.0	40.0	-
Sand 33303	55.0	65.0	-
Brown 383, 30051	8.0	20.0	-
Tan 686A, 33446	62.0	72.0	-
Black 37030	0.0	15.0	-
Aircraft green 34031	-	7.0	-
Interior aircraft black 37031	-	7.0	-
Aircraft gray 36300	-	15.0	-
Dark standstone 33510	-	45.0	-

1/ Color chip number conforms to FED-STD-595.
2/ For wavelength definition, see table II.
3/ The ratio is calculated by dividing the value of the infrared by the value of the red spectral range.

TABLE II. Selected ordinates for determining infrared and red reflectance values from spectrophotometric ordinates.

Magenta red region nanometers (nm)	Infrared region (nm)		
620.0	714.0	769.0	816.0
626.0	725.0	773.0	821.0
638.0	730.0	777.0	826.0
645.0	737.0	783.0	831.0
649.0	742.0	787.0	836.0
652.0	747.0	793.0	842.0
653.0	751.0	797.0	848.0
655.0	756.0	802.0	855.0
658.0	760.0	807.0	862.0
663.0	764.0	811.0	873.0

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3.3.2 Color. The color of the cover material shall be as specified in the contract (see 4.4.3.2 and 6.2).

3.3.3 Glossiness. There shall be low gloss on both sides (Class A) of the cover material or one side only (Class B) (see 6.2). The gloss reading at 60 degrees shall be not greater than 1.5 units and, at 85 degrees, shall be not greater than 3.5 units (see 4.4.3.3). The cover material shall have low gloss on the same side as infrared reflectance (see 3.3.1).

3.3.4 Dimensional stability. The cover material shall show no tendency to curl at the edges or wrinkle when lying flat, and shall have a maximum shrinkage/growth of 0.4 percent in both the warp and fill directions (see 4.4.3.4).

3.3.5 Breaking force.

3.3.5.1 Base cloth. The base cloth of the cover material shall have a minimum break force of 250 pounds (lb) in both the warp and the fill direction. (see 4.4.3.5.1).

3.3.5.2 Cover material. The cover material shall have a minimum break force of 325 lb in the warp direction and 250 lb in the fill direction, and the percent elongation at break shall be within a range of 15 to 40 percent in the warp direction and a range of 25 to 50 percent in the fill direction (see 4.4.3.5.2).

3.3.6 Breaking force after accelerated weathering. The cover material shall have a minimum break force of 275 lb in the warp direction and 180 lb in the fill direction after exposure to 250 hours of accelerated weathering (see 4.4.3.6).

3.3.7 Tearing strength. The cover material shall have a minimum tear strength of 50 lb in both the warp and fill directions (see 4.4.3.7).

3.3.8 Hydrostatic resistance. The cover material shall exhibit no leakage when subjected to a static pressure head of 100 centimeters (cm) for 1 hour at 140 degrees Fahrenheit (°F). The cover material shall meet this requirement in both the body of the fabric as well as the seam areas (see 4.4.3.8).

3.3.9 Coating adhesion. Both sides of the cover material shall have a minimum coating adhesion of 20 lb per 2 inches (in.) width in the warp direction (see 4.4.3.9).

3.3.10 Stiffness. The cover material shall have maximum stiffness in the warp direction of 16.5 cm and 22 cm deflection at 70°F and 10°F, respectively (see 4.4.3.10).

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3.3.11 Blocking resistance at elevated temperature. The cover material shall have a blocking rating of no greater than 2, in accordance with ASTM D751, after 1 hour of exposure at 180°F ± 2°F (see 4.4.3.11).

3.3.12 Resistance to accelerated aging. The cover material shall not experience color change, nor increased softness, stiffness, tackiness, or brittleness when exposed to accelerated aging (see 4.4.3.12).

3.3.13 Resistance to aromatic hydrocarbon fluids. The cover material shall not crack or flake when creased sharply on itself in both the warp and fill directions when immersed in Reference Fuel D (see 4.4.3.13).

3.3.14 Abrasion resistance. The cover material shall meet the requirements as specified in ASTM D3389 using H-18 abrasive wheels and 1000 gram weights. Samples shall show a minimum value of 300 cycles on the face and 100 cycles on the back (see 4.4.3.14).

3.4 Interface requirements.

3.4.1 Weight. The cover material shall have a minimum weight of 19 ounces per square yard (oz/yd^2) and a maximum weight of 22 oz/yd^2 (see 4.4.4.1).

3.5 Support and ownership requirements.

3.5.1 Hazardous materials. The cover material shall not produce allergenic reactions, including but not limited to, dermal sensitization, dermal irritation and eye irritation to military personnel handling the material (see 4.4.5.1).

3.5.2 Odor. The cover material odor rating shall not exceed a rating number of 2 in accordance with SAE J1351 and no toxic fumes shall be present when subjected to 140°F for six hours minimum under wet and dry conditions (see 4.4.5.2).

3.6 Environmental requirements.

3.6.1 Resistance to low/high temperatures. The cover material shall provide acceptable uninterrupted performance, without deterioration, under any extreme condition of inclement weather and in ambient air temperatures from -50 °F to +125°F. The performance of the cover material shall not deteriorate at temperature extremes for exposure duration of up to 6 months. Additionally, the cover material shall be capable of being stored, without degradation of performance, for two years at a time at temperatures that may cycle daily to 140°F (see 4.4.6.1 and 4.4.6.2).

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3.6.2 Mildew resistance. The cover material shall not exhibit growth of mildew or fungus (0 rating), when subjected to a fungi laden atmosphere at 82 - 86°F, at not less than 85% relative humidity for 28 days (see 4.4.6.3).

3.6.3 Flame resistance. The cover material shall not permit flame progression of more than 4 inches per minute (in./min) (see 4.4.6.4).

3.6.4 Oil resistance. Lubricating oil conforming to SAE Viscosity Grade 50 of SAE J1966 shall not leak through, seep through, or permeate the cover material (see 4.4.6.5).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2 and 6.3), first article inspection shall consist of the first article tests specified in table III. First article samples shall be as specified in the contract (see 6.2 and 6.3).

4.3 Conformance inspection. Unless otherwise specified (see 6.2 and 6.4), a sample selected in accordance with 4.3.1 shall be subjected to the conformance tests and examinations specified in table III.

4.3.1 Sampling. Samples from an inspection lot shall be selected in accordance with the sampling plan as specified in the contract (see 6.2 and 6.4). Noncompliance with any specified requirement shall be cause for rejection.

4.4 Verification methods. The types of verification methods included in this section are visual, inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.4.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives the replace verifications required by this specification.

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4.4.2 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, certifications, industry standards, test reports, and rating data.

Table III. Verification methods.

Title	Requirements	Inspections	First article inspection	Conformance inspection
Materials	3.2	4.4.2	X	X
Operating requirements	3.3	4.4.3		
Infrared reflectance	3.3.1	4.4.3.1	X	X
Color	3.3.2	4.4.3.2	X	X
Glossiness	3.3.3	4.4.3.3	X	X
Dimensional stability	3.3.4	4.4.3.4	X	X
Breaking force	3.3.5	4.4.3.5		
Base cloth	3.3.5.1	4.4.3.5.1	X	X
Cover material	3.3.5.2	4.4.3.5.2	X	X
Breaking force after accelerated weathering	3.3.6	4.4.3.6	X	X
Tearing strength	3.3.7	4.4.3.7	X	X
Hydrostatic resistance	3.3.8	4.4.3.8	X	X
Coating adhesion	3.3.9	4.4.3.9	X	X
Stiffness	3.3.10	4.4.3.10	X	X
Blocking resistance at elevated temperature	3.3.11	4.4.3.11	X	X
Resistance to accelerated aging	3.3.12	4.4.3.12	X	X
Resistance to aromatic hydrocarbon fluids	3.3.13	4.4.3.13	X	X
Abrasion resistance	3.3.14	4.4.3.14	X	X
Interface requirements	3.4	4.4.4		
Weight	3.4.1	4.4.4.1	X	X
Support and ownership requirements	3.5	4.4.5		
Hazardous materials	3.5.1	4.4.5.1	X	X
Odor	3.5.2	4.4.5.2	X	X
Environmental requirements	3.6	4.4.6		
Resistance to low/high temperatures	3.6.1	4.4.6.1, 4.4.6.2	X	X
Mildew resistance	3.6.2	4.4.6.3	X	X
Flame resistance	3.6.3	4.4.6.4	X	X
Oil resistance	3.6.4	4.4.6.5	X	X

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4.4.3 Operating requirements verification.

4.4.3.1 Infrared reflectance. To determine conformance to 3.3.1, the infrared reflectance shall be determined using one or more of the methods specified in 4.4 or 4.4.1, and shall fall within the minimum and maximum values as specified in table I.

4.4.3.2 Color. To determine conformance to 3.3.2, the color shall be visually inspected to verify compliance with the contract requirements (see 6.2).

4.4.3.3 Glossiness. To determine conformance to 3.3.3, the cover material shall be tested in accordance with ASTM D523 and shall exhibit a gloss reading of not greater than 1.5 units at less than 60 degrees, and not greater than 3.5 units at more than 85 degrees.

4.4.3.4 Dimensional stability. To determine conformance to 3.3.4, the cover material shall be tested in accordance with SAE J883 and shall show no tendency to curl at the edges or wrinkle when lying flat, and shall exhibit a maximum shrinkage/growth of 0.4 percent in both the warp and fill directions.

4.4.3.5 Breaking force.

4.4.3.5.1 Base cloth. To determine conformance to 3.3.5.1, the base cloth of the cover material shall be tested in accordance with ASTM D5034, and shall exhibit a minimum break force of 250 lb in both the warp and the fill direction.

4.4.3.5.2 Cover material. To determine conformance to 3.3.5.2, the cover material shall be tested in accordance with ASTM D5035, Test Method 1C-E. The cover material shall exhibit a minimum break force of 325 lb in the warp direction and 250 lb in the fill direction and the percent elongation at break shall be within a range of 15 to 40 percent in the warp direction and a range of 25 to 50 percent in the fill direction.

4.4.3.6 Breaking force after accelerated weathering. To determine conformance to 3.3.6, the cover material shall be exposed to 250 hours of accelerated weathering in accordance with ASTM G26, Test Method 3, followed by 24 hours conditioning at 70°F $\pm$ 2°F and 65 $\pm$ 2 percent relative humidity. After accelerated weathering, the cover material shall then be exposed for two hours at 180°F $\pm$ 5°F, followed by 40 hours at 70°F $\pm$ 2°F and 65 $\pm$ 2 percent relative humidity. The cover material shall then be tested in accordance with ASTM D5035, Test Method 1C-E, and shall exhibit a breaking strength of at least 275 lb in the warp direction and at least 180 lb in the fill direction.

4.4.3.7 Tearing strength. To determine conformance to 3.3.7, the cover material shall be tested in accordance with ASTM D751, Tongue Tear Test Method, using a fixed rate of extension of 12 in. minimum. The tearing strength shall be the average of the five highest peak

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loads of resistance registered during the separation of the tear. The cover material shall exhibit an average tearing strength of not less than 50 lb in both the warp and the fill directions.

4.4.3.8 Hydrostatic resistance. To determine conformance to 3.3.8, the cover material shall be tested in accordance with hydrostatic resistance, Procedure B, Procedure 2 of ASTM D751, and shall exhibit no leakage when subjected to a static pressure head of 100 cm for 1 hour at 140 degrees Fahrenheit (°F). The cover material shall meet this requirement in both the body of the fabric as well as the seam areas.

4.4.3.9 Coating adhesion. To determine conformance to 3.3.9, the cover material shall be tested in accordance with ASTM D751 at a pulling clamp speed of 5 millimeters per second (mm/sec). Both sides of the cover material shall exhibit a minimum coating adhesion of 20 lb per 2 in. width in the warp direction.

4.4.3.10 Stiffness. To determine conformance to 3.3.10, the cover material shall be tested in accordance with TAPPI T451 CM, Preferred Procedure 1, in still air, with the specimens conditioned at the test temperatures for a minimum of 4 hours before testing. The cover material shall exhibit a maximum stiffness in the warp direction of 16.5 cm and 22 cm deflection at 70°F and 10°F, respectively.

4.4.3.11 Blocking resistance at elevated temperature. To determine conformance to 3.3.11, the cover material shall be tested in accordance with ASTM D751, and shall exhibit a blocking rating of no greater than 2 after 1 hour of exposure at 180°F $\pm$ 2°F.

4.4.3.12 Resistance to accelerated aging. To determine conformance to 3.3.12, the cover material shall be tested for breaking strength in accordance with FED-STD-191 or equivalent (see 4.4.1), Method 5852. Additionally, exposed samples shall be evaluated for color change in accordance with AATCC Evaluation Procedure 1 with a minimum rating of 4-5. Stiffness shall be evaluated in accordance with 4.4.3.10. No softness or tackiness shall be observed. Brittleness shall be evaluated by the percent elongation at break, which may be obtained from the break strength test procedure. A breaking elongation after aging of less than 10 percent for the warp direction and 20 percent for the fill direction is unacceptable.

4.4.3.13 Resistance to aromatic hydrocarbon fluids. To determine conformance to 3.3.13, the cover material shall be immersed in Reference Fuel D conforming to ASTM D471 for 5 minutes and then shall be allowed to dry at room temperature for 2 hours $\pm$ 5 minutes. The sample for immersion shall be at least 6 in. by 6 in. in order to minimize edge effects of penetration of the fuel. Evaluation shall be in the center of the sample, and any cracking or flaking after creasing in both the warp and fill directions shall constitute failure of this test.

4.4.3.14 Abrasion resistance. To determine conformance to 3.3.14, the cover material shall be tested in accordance with ASTM D3389, Method A using H-18 abrasive wheels and

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1000 gram weights. Samples shall exhibit a minimum value of 300 cycles on the face and 100 cycles on the back.

4.4.4 Interface requirements.

4.4.4.1 Weight. To determine conformance to 3.4.1, a known size of the cover material shall be weighed and the mass per unit area determined. The weight shall be within 19-22 oz/yd^2.

4.4.5 Support and ownership requirements verification.

4.4.5.1 Hazardous material test. To determine conformance to 3.5.1, the cover material shall be tested in accordance with 40 CFR, Sections 798.4100, 798.4470, and 798.4500, and shall not produce allergenic reactions, including but not limited to, dermal sensitization, dermal irritation and eye irritation.

4.4.5.2 Odor. To determine conformance to 3.5.2, the cover material shall be tested in accordance with SAE J1351, and shall exhibit an odor rating not to exceed a rating number of 2 and no toxic fumes shall be present when subjected to 140°F for 6 hours minimum under wet and dry conditions

4.4.6 Operating environmental requirements.

4.4.6.1 Low temperature bend test. To determine conformance to 3.6.1, the cover material shall rate a “pass“ when tested in accordance with ASTM D2136 at -50°F $\pm$ 5°F for a minimum of 6 hours. Additionally, use one or more of the methods specified in 4.4 or 4.4.1 to verify that the cover material is capable of being stored, without degradation in performance, for two years at storage temperatures that cycle daily to 140°F.

4.4.6.2 High temperature deformation test. To determine conformance to 3.6.1, the cover material shall be exposed to a temperature of 170°F $\pm$ 2°F for a minimum of 6 hours, and then while still maintained at 170°F $\pm$ 2°F, shall withstand a 180° bend over a 0.125 in. diameter steel rod with no cracking, delamination, tackiness, or blistering. At least five specimens, 50 millimeters (mm) x 150 mm, shall be tested in both the warp and fill directions. Samples shall be cut according to the guidelines in ASTM D5035. Specimens shall be rated on a pass/fail basis. A “fail” rating for one or more of the specimens shall be considered a failure of the test.

4.4.6.3 Mildew resistance. To determine conformance to 3.6.2, the cover material shall be tested in accordance with ASTM G21 and shall pass with a rating of zero, when subjected to a fungi laden atmosphere at 82 - 86°F, at not less than 85% relative humidity for 28 days.

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4.4.6.4 Flame resistance. To determine conformance to 3.6.3, the cover material shall be tested in accordance with 49 CFR, Section 571.302, and shall exhibit a flame progression of not more than 4 in./min.

4.4.6.5 Oil resistance. To determine conformance to 3.6.4, a 0.5 in. circle of Viscosity Grade 50 lubricating oil (SAE J1966) shall be poured onto an area of approximately one square inch on one side of the cover material and shall be allowed to stand for one hour. The lubricating oil shall not leak through, seep through, or permeate the cover material.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The cover material covered by this specification is intended for use on the cab and/or cargo compartment of military combat and tactical vehicles. Due to the infrared reflectance requirements and the extreme environmental conditions under which the cover material must perform, this item is military unique.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article inspection is required (see 3.1).
d. Designate whether infrared reflectance and glossiness requirements apply to both sides or one side only (see 3.3.1 and 3.3.3).
e. Color of the coated vehicle cloth (see 3.3.2 and 4.4.3.2).
f. If first article testing is other than as specified (see 4.2).
g. Specification of first article sampling plan (see 4.2).

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h. If conformance testing is other than as specified (see 4.3).
i. Specification of conformance testing sampling plan (see 4.3.1).
j. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first article production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of the factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table I and their designated frequency based on a risk assessment for the procurement.

6.5 Natural fibers. Natural fibers refer to those derived directly from plants. Examples include but are not limited to cotton, hemp, flax, and jute.

6.6 Historical material requirements. Historically, cover base cloth of high-tenacity polyester and a minimum of 25 yarns per inch in both the fill and warp direction or a material with a minimum strength of 250 lb in both the warp and fill direction, both coated with poly (vinyl chloride) (PVC) or polyurethane, have provided the best results as opposed to knitted or less dense materials. Field experience has shown that cover material manufactured from natural fibers has passed first article tests, but perform poorly in the field. The high failure rate and replacement costs associated with natural fiber covers prohibit this as an acceptable base material.

6.6.1 Plain woven materials. Plain woven materials have been specified historically to ensure adequate and uniform seam strength in coated vehicle cover materials. Plain wovens are also among the strongest and most stable forms of fabric weaving. Due to the “lock down” phenomenon of plain wovens, any long-term fabric shifting and skewing is minimized.

6.6.2 High-tenacity polyester. High-tenacity polyester was tested by TACOM in 1994 and found to be the most cost-effective base material available on the market at that time for the production of durable, long-lasting coated vehicle cover materials.

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6.6.3 Coated cloth weight. Historically, a coated cloth weight of 20 ± 1 oz/yd^2 and a minimum of 25 yarns per inch in both the warp and fill directions have served the U.S. Army well, and is recommended here to ensure long term cover material resilience and durability. (This specification is similar to the Canadian and German Army requirements.) The number of yarns per inch is representative of fabric density and strength, as greater numbers of yarns per inch ensure a larger safety cushion in durability should yarns break during use or contain any manufacturing imperfections. The number of yarns per inch also ensures increased coating adhesion which is very important during heat welding operations.

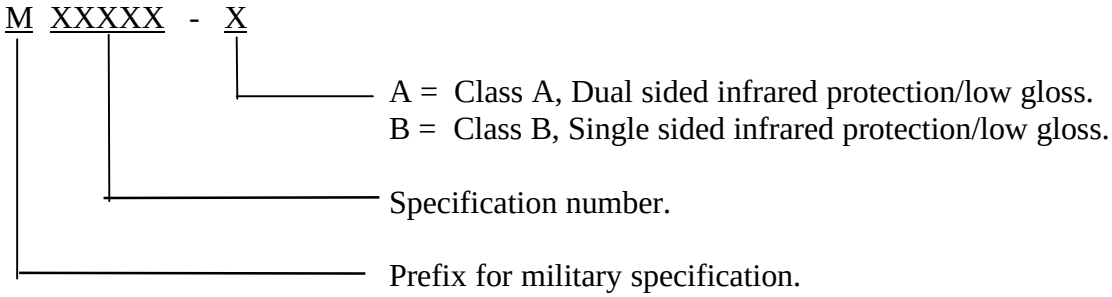
6.6.4 PVC coating. Historically, the chemical characteristics of 14 ± 1.2 oz/yd^2 of PVC coating have ensured long term coating durability and limited plasticizer migration and coating cracking.

6.7 Final vehicle cover assembly. The following information should be considered when a final vehicle cover assembly is constructed.

6.7.1 Grommets. Historically, mechanically-fastened metal grommets have been chosen for load-bearing sections of vehicle cover applications because of their long term durability. As metal grommets are normally mechanically fastened all the way through the vehicle cover, as opposed to plastic grommets, which in the case of heat welding, are fastened to the coating on one side only, metal grommets can better prevent tear-through failures at high-stress, load-bearing sections of the vehicle cover, including cover tie-downs. Metal grommets are also tougher and yield under higher stresses than plastic grommets. Furthermore, plastic grommets are adversely affected in the extreme temperature range the vehicle cover may encounter.

6.7.2 Seam strength. The breaking force of load-bearing seams should be the same as that for the coated cloth.

6.8 Part or Identifying Number (PIN). The PINs to be used for cover materials acquired to this specification are created as shown below.



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6.9 Subject term (key word) listing.

Cab
Cargo
Fill
Flame resistance
Mildew
Warp
Weather resistant

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - YD1
Air Force - 99

Preparing Activity:
Army - AT

(Project 25GP-0011)

Review Activities:
Air Force - 84
DLA – CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document letter should be
2. The submitter of this form must complete blocks 4,
3. The preparing activity must provide a reply within 30 days from

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
	MIL-PRF-32002B	980918

3. DOCUMENT TITLE
COVER MATERIAL, VEHICLE, COATED TACTICAL AND COMBAT, HEAVY-DUTY, WATERPROOF

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*
	(1) Commercial (810) 574-8745 (2) AUTOVON 786-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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